

This assignment calls for enhancing the program that you wrote for homework #8. You are to add two new items to the menu used in homework #8, namely ReadList and SortList. The new menu looks like:

A(ddNode B(uildList D(eleteNode R(eadList S(ortList Z(apList Q(uit =>

The menu option "R(eadList" causes TestReadList to be called, which in turn calls ReadList, a function that reads characters from a file and stores them in the linked list. The function ReadList is documented below. **Note that ReadList should read all chars**, including newline characters until eof is reached. **Do not use AddNode in ReadList!**

```
/* ***** ReadList *****
DESCRIPTION  Builds a singly linked list with a dummy head node. The
             characters in the list are read in from an external file
             in the same order in which they are found in file.

             Input to list terminates when the End of File is encountered
PARAMETERS
IN : L       A reference to a singly linked list with a dummy head node.
             It is imperative that List be initialized before calling
             this routine.
IN : FileName A string that has the name of the file to open, if error
             in opening then return a false.
RETURNS:     true if file opened successfully, else false

NOTE        Before building the new list, ZapList is called.
-----*/
```

The menu option "S(ortList" causes TestSortList to be called, which in turn causes SortList to be called. SortList rearranges the linked list so that it is in natural order, i.e. sorted. Below are the header comments that describe the SortList function.

```
/* ***** SortList *****
Description Arranges the singly linked list pointed to by L in
             natural order. It is assumed that the list has a dummy head node.

             The algorithm used is a linked variation of the selection
             sort and works like this:

             Start with EndSorted aimed at first node of list
             repeat
             Find smallest char between EndSorted and end of list
             Swap smallest element with char in EndSorted
             Change EndSorted to next node
             until we get to end of list

             None of the references in linked list are changed

Parameters
IN : L       A reference to a singly linked list with a dummy head node
-----*/
```

Notes/Comments

- * Do not code the Bubble sort algorithm -- read comments above!
- * Do not re-test the routines that you finished when doing homework #8, but yet you will have to build a list to test the Sortlist.
- * In testing ReadList, you should not read in large files. I suggest that you test it on the files, words.1 through words.5, that were used in homework #4 and earlier ones.
- * The ReadList function is basically doing like was done in homework #4 with file input.
- * This assignment will be run from the IDE and NOT the command line.
- * Turn in only your output, ReadList function, and the SortList function. (points off if you turn in more).
- * Program shell can be found on Canvas as "a9shell.java", which you must use. Will need to fill in the missing functions from homework #8 as well.